

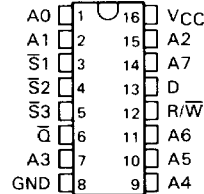
SN74S201, SN74S301 256-BIT HIGH-PERFORMANCE RANDOM-ACCESS MEMORIES

D2007, SEPTEMBER 1977—FEBRUARY 1984

STATIC RANDOM-ACCESS MEMORIES

- Static Fully Decoded RAM's Organized as 256 Words of One Bit Each
- Schottky-Clamped for High Performance
- Choice of Three-State or Open-Collector Outputs
- Compatible with Most TTL and I²L Circuits
- Chip-Select Input Simplify External Decoding
- Typical Performance:
Read Access Time . . . 42 ns
Power dissipation . . . 500 mW

SN74S201, SN74S301 . . . J OR N PACKAGE
(TOP VIEW)



description

These 256-bit active-element memories are monolithic transistor-transistor logic (TTL) arrays organized as 256 words of one bit. They are fully decoded and have three chip-select inputs to simplify decoding required to achieve expanded system organizations.

write cycle

The information applied at the data input is written into the selected location when the chip-select inputs and the write-enable input are low. While the write-enable input is low, the 'S201 outputs are in the high-impedance state and the 'S301 outputs are off. When a number of outputs are bus-connected, this high-impedance or off state will neither load nor drive the bus line, but it will allow the bus line to be driven by another active output or a passive pull-up.

read cycle

The stored information (complement of information applied at the data input during the write cycle) is available at the output when the write-enable input is high and the three chip-select inputs is low. When any one of the chip-select inputs are high, the 'S201 outputs will be in the high-impedance state and the 'S301 outputs will be off.

FUNCTION TABLE

FUNCTION	INPUTS		'S201 OUTPUT ($\bar{Q}$)	'S301 OUTPUT ($\bar{Q}$)
	CHIP SELECT $\bar{S}$	WRITE ENABLE R/W		
Write	L	L	High Impedance	Off
Read	L	H	Complement of Data Entered	Complement of Data Entered
Inhibit	H	X	High Impedance	Off

H = high level, L = low level, X = irrelevant

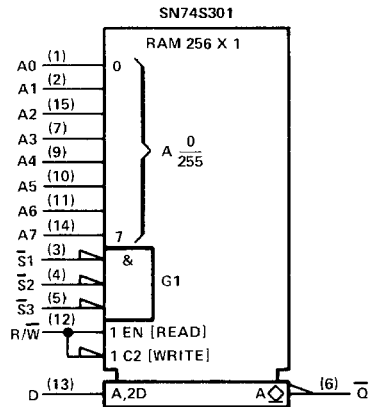
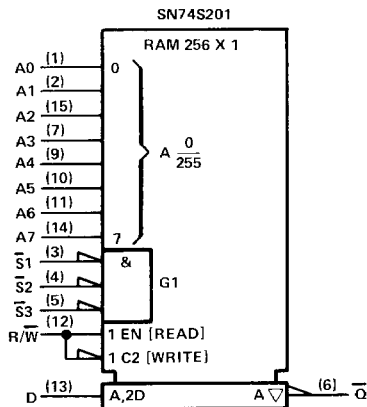
For chip-select: L = all $\bar{S}_i$ inputs low, H = one or more $\bar{S}_i$ inputs high

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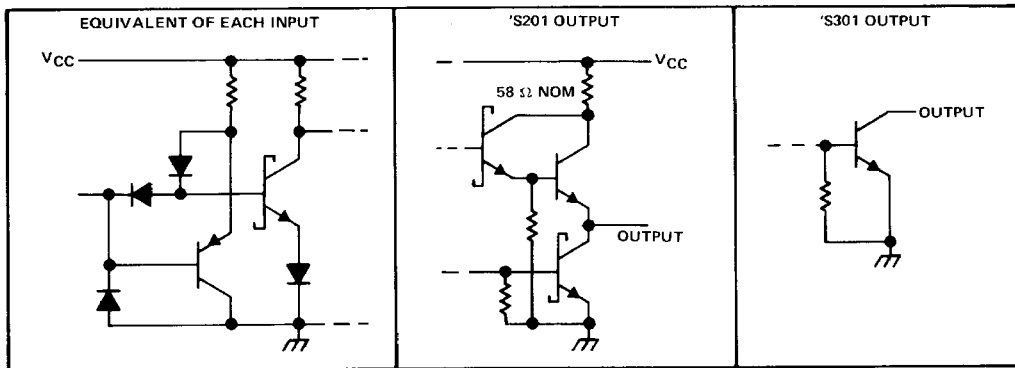
RAMs

SN74S201, SN74S301 256-BIT HIGH-PERFORMANCE RANDOM-ACCESS MEMORIES

logic symbols



schematics of inputs and outputs



absolute maximum ratings over operating free-air temperature range (unless otherwise noted)

Supply voltage, V_{CC} (see Note 1)	7 V
Input voltage	5.5 V
Off-State output voltage	5.5 V
Operating free-air temperature range	0°C to 70°C
Storage temperature range	-65°C to 150°C

SN74S201, SN74S301 256-BIT HIGH-PERFORMANCE RANDOM-ACCESS MEMORIES

recommended operating conditions

		SN74S201			SN74S301			UNIT
		MIN	NOM	MAX	MIN	NOM	MAX	
Supply voltage, V_{CC} (see Note 1)		4.75	5	5.25	4.75	5	5.25	V
High-level output voltage, V_{OH}						5.5		V
High-level output current, I_{OH}				-10.3				mA
Low-level output current, I_{OL}				16		16		mA
Width of write pulse (write enable low), $t_W(wr)$		65			65			ns
Setup time	Address before write pulse, $t_{su}(ad)$	0†			0†			ns
	Data before end of write pulse, $t_{su}(da)$	65†			65†			
	Chip select before end of write pulse, $t_{su}(\bar{S})$	65†			65†			
Hold time	Address after write pulse, $t_h(ad)$	0†			0†			ns
	Data after write pulse, $t_h(da)$	0†			0†			
	Chip select after write pulse, $t_h(\bar{S})$	0†			0†			
Operating free-air temperature, T_A		0	70		0	70		°C

† The arrow indicates the transition of the write-enable input used for reference: ↓ for the low-to-high transition, ↓ for the high-to-low transition.
NOTE 1: Voltage values are with respect to network ground terminal.

electrical characteristics over recommended operating free-air temperature range (unless otherwise noted)

PARAMETER		TEST CONDITIONS†	'S201			'S301			UNIT
			MIN	TYP‡	MAX	MIN	TYP‡	MAX	
V_{IH}	High-level input voltage		2			2			V
V_{IL}	Low-level input voltage				0.8			0.8	V
V_{IK}	Input clamp voltage	$V_{CC} = \text{MIN}, I_I = -18 \text{ mA}$			-1.2			-1.2	V
V_{OH}	High-level output voltage	$V_{CC} = \text{MIN}, V_{IH} = 2 \text{ V}, V_{IL} = 0.8 \text{ V}, I_{OH} = \text{MAX}$	2.4						V
V_{OL}	Low-level output voltage	$V_{CC} = \text{MIN}, V_{IH} = 2 \text{ V}, I_{OL} = 16 \text{ mA}$		0.45			0.45		V
I_{OH}	High-level output current	$V_{CC} = \text{MIN}, V_{IH} = 2 \text{ V}, V_O = 2.4 \text{ V}$					40		μA
		$V_{IL} = 0.8 \text{ V}, V_O = 5.5 \text{ V}$					100		
I_{OZH}	Off-state output current, high-level voltage applied	$V_{CC} = \text{MAX}, V_{IH} = 2 \text{ V}, V_{IL} = 0.8 \text{ V}, V_{OH} = 2.4 \text{ V}$		40					μA
I_{OZL}	Off-state output current, low-level voltage applied	$V_{CC} = \text{MAX}, V_{IH} = 2 \text{ V}, V_{IL} = 0.8 \text{ V}, V_{OL} = 0.5 \text{ V}$		-40					μA
I_I	Input current at maximum input voltage	$V_{CC} = \text{MAX}, V_I = 5.5 \text{ V}$		1			1		mA
I_{IH}	High-level input current	$V_{CC} = \text{MAX}, V_I = 2.7 \text{ V}$		25			25		μA
I_{IL}	Low-level input current	$V_{CC} = \text{MAX}, V_I = 0.5 \text{ V}$		-250			-250		μA
I_{OS}	Short-circuit output current§	$V_{CC} = \text{MAX}$	-30		-100				mA
I_{CC}	Supply current	$V_{CC} = \text{MAX}, \text{ See Note 2}$	100	140		100	140		mA

† For conditions shown as MIN or MAX use the appropriate value specified under recommended operating conditions.

‡ All typical values are at $V_{CC} = 5 \text{ V}, T_A = 25^\circ\text{C}$.

§ Duration of the short circuit should not exceed one second.

NOTE 2: I_{CC} is measured with all chip select inputs grounded, all other inputs at 4.5 V, and the output open.

5
RAMs

SN74S201, SN74S301
256-BIT HIGH-PERFORMANCE RANDOM-ACCESS MEMORIES

'S201 switching characteristics over recommended operating ranges of T_A and V_{CC}
(unless otherwise noted)

PARAMETER		TEST CONDITIONS	MIN	TYP [†]	MAX	UNIT
$t_{a(ad)}$	Access time from address	$C_L = 30 \text{ pF}$, See Note 3		42	65	ns
$t_{a(S)}$	Access time from chip select (select time)			13	30	ns
t_{SR}	Sense recovery time			20	40	ns
t_{PXZ}	Disable time from high or low level	From $\bar{S}$		9	20	ns
		From $R/\bar{W}$				

'S301 switching characteristics over recommended operating ranges of T_A and V_{CC}
(unless otherwise noted)

PARAMETER			TEST CONDITIONS	MIN	TYP [‡]	MAX	UNIT
$t_{a(ad)}$	Access time from address		$C_L = 30 \text{ pF}$, $R_{L1} = 300 \, \Omega$, $R_{L2} = 600 \, \Omega$, See Note 3		42	65	ns
$t_{a(S)}$	Access time from chip enable (enable time)				13	30	ns
t_{SR}	Sense recovery time				20	40	ns
t_{PLH}	Propagation delay time, low-to-high-level	From $\overline{S}$		8	20	ns	
	output (disable time)	From $R/\overline{W}$		15	35		

[†]All typical values are at $V_{CC} = 5 \text{ V}$, $T_A = 25^\circ$.

NOTE 2: Load circuits and voltage waveforms are shown in Section 1.

5

RAMS